

1. (5+5+5+5 pts) Select the correct answer. No justification is required.

I. The value of $\lim_{(x,y) \rightarrow (0,\pi/4)} (x+y) \sin(x-2y)$ is:

- (A) $-\frac{\pi}{2}$ (B) 0 (C) $\frac{\pi}{4} - 1$ (D) $\frac{\pi}{4}$ (E) $-\frac{\pi}{4}$

II. If $\sin(y+z) + \ln x = 3 - z - y$, then $\frac{\partial z}{\partial y} =$

- (A) $-1 - \cos(y+z)$ (B) $-\cos(y+z)$ (C) -1
(D) $1 + \cos(y+z)$ (E) $\cos(y+z)$

III. The gradient of a differentiable function $f(x, y)$ is:

- (A) $f_x dx + f_y dy$ (B) $f_x \mathbf{i} + f_y \mathbf{j}$ (C) $f_x + f_y$
(D) $f_x u_1 + f_y u_2$ (E) $\mathbf{i} + \mathbf{j}$

IV. If $f(x, y, z) = \sinh^3(xyz)$, then $\frac{\partial f}{\partial x} =$

- (A) $3yz \sinh^2(xyz)$ (B) $3yz \cosh^2(xyz)$ (C) $yz \cosh^3(xyz)$
(D) $3yz \sinh^2(xyz) \cosh(xyz)$ (E) $3 \sinh^2(xyz) \cosh(xy^2 z^2)$

2. (10 pts) Find an equation for the tangent plane to the surface $z = \frac{x}{y^2}$ at $(3, 2, 3/4)$.

3. (10 pts) Find the limit, if it exists, or show that the limit does not exist:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \tan(xy)}{x^2 + 5y^2}$$

4. (10 pts) Find and sketch the domain of the function:

$$f(x, y) = \arcsin(x^2 + y^2 - 1)$$

5. (15 pts) Let $f(x, y) = ye^{-xy}$. Find all unit vectors $\mathbf{u}$, such that $D_{\mathbf{u}}f(0, 1) = 1$.

6. (15 pts) Show that $w = \sqrt{x^2 + y^2 + z^2}$ is a solution of the differential equation:

$$w_{xx} + w_{yy} + w_{zz} = \frac{2}{w}$$

7. (20 pts) Find the critical points of the function and classify them as local max/min or saddle points:

$$f(x, y) = x^3 + y^2 - 6xy + 6x + 3y - 2$$

Solutions.

1. I (E) II (C) III (B) IV (D)

2. $f(x,y) = \frac{x}{y^2} \Rightarrow f_x = \frac{1}{y^2} \Rightarrow f_x(3,2) = \frac{1}{4}$
 $f_y = -2xy^{-3} \Rightarrow f_y(3,2) = -3/4$

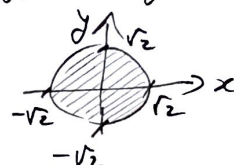
$f(3,2) = \frac{3}{4} \Rightarrow$ T.P. $z = \frac{3}{4} + \frac{1}{4}(x-3) - \frac{3}{4}(y-2)$

3. $x^2 \leq x^2 + 5y^2 \Rightarrow \left| \frac{x^2 \tan(xy)}{x^2 + 5y^2} \right| \leq |\tan(xy)| \rightarrow |\tan 0| = 0$
as $(x,y) \rightarrow (0,0)$

$\Rightarrow$ by Squeeze Th: $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 \tan(xy)}{x^2 + 5y^2} = 0$.

4. $D_f = \{(x,y) : -1 \leq x^2 + y^2 - 1 \leq 1\} = \{(x,y) : x^2 + y^2 \leq 2\}$

Closed disc center $(0,0)$, radius $\sqrt{2}$.



5. $\nabla f = \langle -y^2 e^{-xy}, e^{-xy} - xy e^{-xy} \rangle \Rightarrow \nabla f(0,1) = \langle -1, 1 \rangle$

$\bar{u} = \langle u_1, u_2 \rangle \Rightarrow D_{\bar{u}} f(0,1) = \nabla f \cdot \bar{u} = -u_1 + u_2 = 1 \Rightarrow u_2 = 1 + u_1$

$u_1^2 + u_2^2 = 1 \Rightarrow u_1^2 + (1 + u_1)^2 = 1 \Rightarrow u_1 = 0 \text{ or } -1 \Rightarrow \bar{u} = \langle 0, 1 \rangle$
or $\langle -1, 0 \rangle$.

6. $w_x = x(x^2 + y^2 + z^2)^{-1/2}$

$\Rightarrow w_{xx} = (x^2 + y^2 + z^2)^{-1/2} - x^2(x^2 + y^2 + z^2)^{-3/2} = \frac{1}{w} - \frac{x^2}{w^3}$

Similarly, $w_{yy} = \frac{1}{w} - \frac{y^2}{w^3}$ and $w_{zz} = \frac{1}{w} - \frac{z^2}{w^3}$

$\Rightarrow w_{xx} + w_{yy} + w_{zz} = \frac{3}{w} - \frac{x^2 + y^2 + z^2}{w^3} = \frac{3}{w} - \frac{1}{w} = \frac{2}{w}$.

7. C.P. $\begin{cases} f_x = 3x^2 - 6y + 6 = 0 & (1) \\ f_y = 2y - 6x + 3 = 0 \Rightarrow y = 3x - \frac{3}{2} & (2) \end{cases}$

Subs. (2) in (1): $3x^2 - 18x + 15 = 0 \Rightarrow x^2 - 6x + 5 = 0 \Rightarrow (x-1)(x-5) = 0$

$\Rightarrow x = 1 \text{ or } 5 \Rightarrow 2 \text{ C.P. } (1, \frac{3}{2}) \text{ \& } (5, \frac{27}{2})$.

$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = \begin{vmatrix} 6x & -6 \\ -6 & 2 \end{vmatrix} = 12(x-3)$

$D(1, \frac{3}{2}) = -24 < 0 \Rightarrow$ saddle point at $(1, \frac{3}{2})$

$D(5, \frac{27}{2}) = 24 > 0$ & $f_{yy} = 2 > 0 \Rightarrow$ local min at $(5, \frac{27}{2})$.